# T3 intermediate-EFT one-loop RGE term comparison: EFT_1 after $F$

This stage has active field content SM+$S_1 + S_2$. The report combines the renormalisable RGBeta running with the dimension-five $\psi^2\phi^2$ Wilson-coefficient running generated after integrating out $F$. Throughout, $16\pi^2\,\mu\,dX/d\mu = \beta_X^{(1)}$. For the Wilson sector only the tree-generated stage-1 coefficient $C^{(0)}$ is inserted into the one-loop anomalous dimension; the one-loop threshold piece is retained as a boundary term and is not run again at this order.

## Notation key

| symbol | interaction / definition |
|---|---|
| $g_1$ | $D_\mu \supset i g_1 Y B_\mu$ |
| $g_2$ | $D_\mu \supset i g_2 T^A W_\mu^A$ |
| $g_3$ | $D_\mu \supset i g_3 t^A G_\mu^A$ |
| $Y_u$ | $-\bar{Q} Y_u \widetilde{H} u + \text{h.c.}$ |
| $Y_d$ | $-\bar{Q} Y_d H d + \text{h.c.}$ |
| $Y_e$ | $-\bar{L} Y_e H e + \text{h.c.}$ |
| $y_1$ | $L F S_1 + \text{h.c.}$ |
| $y_2$ | $L F S_2 + \text{h.c.}$ |
| $M_F$ | $-\bar{F} M_F F$ |
| $T$ | $\mathrm{Tr}(Y_e^\dagger Y_e + 3 Y_u^\dagger Y_u + 3 Y_d^\dagger Y_d)$ |
| $T_\nu^{(1)}$ | $\mathrm{Tr}(y_1^\dagger y_1)$ |
| $T_\nu^{(2)}$ | $\mathrm{Tr}(y_2^\dagger y_2)$ |
| $T_\nu$ | $T_\nu^{(1)} + T_\nu^{(2)}$ |
| $m_1^2$ | $m_1^2 S_1^\dagger S_1$ |
| $m_2^2$ | $m_2^2 S_2^\dagger S_2$ |
| $\lambda_1$ | $\frac{1}{2}\lambda_1 (H^\dagger H)^2$ |
| $\lambda_{S_1}^{(1)}$ | $\frac{1}{2}\lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$ |
| $\lambda_{S_2}^{(1)}$ | $\frac{1}{2}\lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$ |
| $\lambda_{HS_1}^{(1)}$ | $\lambda_{HS_1}^{(1)} (H^\dagger H)(S_1^\dagger S_1)$ |
| $\lambda_{HS_2}^{(1)}$ | $\lambda_{HS_2}^{(1)} (H^\dagger H)(S_2^\dagger S_2)$ |
| $\lambda_{HS_1}^{(A)}$ | $\lambda_{HS_1}^{(A)} (H^\dagger T^A H)(S_1^\dagger T^A S_1)$ |
| $\lambda_{HS_2}^{(A)}$ | $\lambda_{HS_2}^{(A)} (H^\dagger T^A H)(S_2^\dagger T^A S_2)$ |
| $\lambda_5$ | $\lambda_5 H H S_1 S_2^\dagger + \text{h.c.}$ |
| $\lambda_{12}^{(1)}$ | $\lambda_{12}^{(1)} (S_1^\dagger S_1)(S_2^\dagger S_2)$ |
| $\lambda_{12}^{(A)}$ | $\lambda_{12}^{(A)} (S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$ |
| $\lambda_{12}^{(\times)}$ | $\lambda_{12}^{(\times)} (S_1^\dagger S_2)(S_2^\dagger S_1)$ |
| $\lambda_{S_1}^{(A)}$ | $\lambda_{S_1}^{(A)} (S_1^\dagger T^A S_1)(S_1^\dagger T^A S_1)$ |
| $\lambda_{S_2}^{(A)}$ | $\lambda_{S_2}^{(A)} (S_2^\dagger T^A S_2)(S_2^\dagger T^A S_2)$ |
| $\alpha$ | $Y_{S_1} = \alpha/2,\ Y_F = (\alpha+1)/2,\ Y_{S_2} = (\alpha+2)/2$ |

| | |
|---|---|
| $C_2(d)$ | $(d^2-1)/4$ |
| $T(d)$ | $d(d^2-1)/12$ |
| $C^{ij;AB}_{LLS_1S_2}$ | $L_iL_jS_1^AS_2^B$ |
| $C_{ijab}$ | $L_iL_j\phi_a\phi_b$ |
| $C_5^{ij}$ | $(L_i^TCL_j)HH$ |
| $R_W$ | $16\pi^2\beta_{C_5} \supset R_W\lambda_5 C_{LLS_1S_2}$ |

## Renormalisable EFT1 couplings

$\beta^{(1)}_{g_1}$

**Quantity:** $g_1$

$$16\pi^2\,\mu\frac{dg_1}{d\mu} = \beta^{(1)}_{g_1}$$

| Column | Representative term structure |
|---|---|
| T1 | $\frac{17{g_1}^3}{2}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $\frac{17{g_1}^3}{2}$ |

$\beta_{g_2}^{(1)}$

**Quantity:** $g_2$

$$16\pi^2\,\mu\frac{dg_2}{d\mu} = \beta_{g_2}^{(1)}$$

| Column | Representative term structure |
|---|---|
| T1 | $-\frac{17{g_2}^3}{6}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $-\frac{17{g_2}^3}{6}$ |

$\beta_{g_3}^{(1)}$

**Quantity:** $g_3$

$$16\pi^2\,\mu\frac{dg_3}{d\mu} = \beta_{g_3}^{(1)}$$

| Column | Representative term structure |
|---|---|
| T1 | $-7{g_3}^3$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $-7{g_3}^3$ |

$\beta^{(1)}_{Y_u}$

**Quantity:** $Y_u$

$$16\pi^2 \, \mu \frac{dY_u}{d\mu} = \beta^{(1)}_{Y_u}$$

# Terms 1 of 3

| Column | Representative term structure |
|---|---|
| T1 | $3 \, \mathrm{Tr}\left(Y_d \ (Y_d)^\dagger\right) (Y_u)_{ij}$ |
| T2 | $+ \, \mathrm{Tr}\left(Y_e \ (Y_e)^\dagger\right) (Y_u)_{ij}$ |
| T3 | $+3 \, \mathrm{Tr}\left(Y_u \ (Y_u)^\dagger\right) (Y_u)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $3 \, \mathrm{Tr}\left(Y_d \ (Y_d)^\dagger\right) (Y_u)_{ij}$ | $+ \, \mathrm{Tr}\left(Y_e \ (Y_e)^\dagger\right) (Y_u)_{ij}$ | $+3 \, \mathrm{Tr}\left(Y_u \ (Y_u)^\dagger\right) (Y_u)_{ij}$ |

**Quantity:** $Y_u$

# Terms 2 of 3

| Column | Representative term structure |
|---|---|
| T1 | $-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$ |
| T2 | $-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$ |
| T3 | $-8{g_3}^2\ (Y_u)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$ | $-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$ | $-8{g_3}^2\ (Y_u)_{ij}$ |

**Quantity:** $Y_u$

# Terms 3 of 3

| Column | Representative term structure |
|---|---|
| T1 | $-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$ |

$\beta^{(1)}_{Y_d}$

**Quantity:** $Y_d$

$$16\pi^2\,\mu\frac{dY_d}{d\mu} = \beta^{(1)}_{Y_d}$$

# Terms 1 of 3

| Column | Representative term structure |
|---|---|
| T1 | $\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$ |
| T2 | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$ |
| T3 | $+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$ |

**Quantity:** $Y_d$

# Terms 2 of 3

| Column | Representative term structure |
|---|---|
| T1 | $+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$ |
| T2 | $-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$ |
| T3 | $-8{g_3}^2\ (Y_d)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$ | $-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$ | $-8{g_3}^2\ (Y_d)_{ij}$ |

**Quantity:** $Y_d$

# Terms 3 of 3

| Column | Representative term structure |
|---|---|
| T1 | $-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$ |

$\beta_{Y_e}^{(1)}$

**Quantity:** $Y_e$

$$16\pi^2\,\mu\frac{dY_e}{d\mu} = \beta_{Y_e}^{(1)}$$

# Terms 1 of 2

| Column | Representative term structure |
|---|---|
| T1 | $3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$ |
| T2 | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$ |
| T3 | $+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$ | $+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$ |

**Quantity:** $Y_e$

# Terms 2 of 2

| Column | Representative term structure |
|---|---|
| T1 | $+\frac{3}{2}\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij}$ |
| T2 | $-\frac{9}{4}{g_2}^2\ (Y_e)_{ij}$ |
| T3 | $-\frac{15}{4}{g_1}^2\ (Y_e)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $+\frac{3}{2}\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij}$ | $-\frac{9}{4}{g_2}^2\ (Y_e)_{ij}$ | $-\frac{15}{4}{g_1}^2\ (Y_e)_{ij}$ |

$\beta^{(1)}_{m_1^2}$

**Quantity:** $m_1^2$

$$16\pi^2\,\mu\frac{dm_1^2}{d\mu} = \beta^{(1)}_{m_1^2}$$

# Terms 1 of 2

| Column | Representative term structure |
|---|---|
| T1 | $-\frac{9{g_2}^2 m_1^2}{2}$ |
| T2 | $-\frac{3{g_1}^2 m_1^2}{2}$ |
| T3 | $+4\lambda^{(1)}_{12} m_2^2$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $-\frac{9{g_2}^2 m_1^2}{2}$ | $-\frac{3{g_1}^2 m_1^2}{2}$ | $+4\lambda^{(1)}_{12} m_2^2$ |

**Quantity:** $m_1^2$

# Terms 2 of 2

| Column | Representative term structure |
|---|---|
| T1 | $+6\lambda_{S_1}^{(1)} m_1^2$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $+6\lambda_{S_1}^{(1)} m_1^2$ |

$\beta^{(1)}_{m_2^2}$

**Quantity:** $m_2^2$

$$16\pi^2\,\mu\frac{dm_2^2}{d\mu} = \beta^{(1)}_{m_2^2}$$

# Terms 1 of 2

| Column | Representative term structure |
|---|---|
| T1 | $-\frac{9{g_2}^2 m_2^2}{2}$ |
| T2 | $-\frac{27{g_1}^2 m_2^2}{2}$ |
| T3 | $+4\lambda^{(1)}_{12} m_1^2$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $-\frac{9{g_2}^2 m_2^2}{2}$ | $-\frac{27{g_1}^2 m_2^2}{2}$ | $+4\lambda^{(1)}_{12} m_1^2$ |

**Quantity:** $m_2^2$

# Terms 2 of 2

| Column | Representative term structure |
|---|---|
| T1 | $+6\lambda_{S_2}^{(1)} m_2^2$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $+6\lambda_{S_2}^{(1)} m_2^2$ |

$\beta^{(1)}_{\lambda_1}$

**Quantity:** $\lambda_1$

$$16\pi^2\,\mu\frac{d\lambda_1}{d\mu} = \beta^{(1)}_{\lambda_1}$$

# Terms 1 of 5

| Column | Representative term structure |
|---|---|
| T1 | $\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ |
| T2 | $-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$ |
| T3 | $-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ | $-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$ | $-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ |

**Quantity:** $\lambda_1$

# Terms 2 of 5

| Column | Representative term structure |
|---|---|
| T1 | $-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$ |
| T2 | $+4\lambda_5\ (\lambda_5)^*$ |
| T3 | $+\frac{9{g_2}^4}{4}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$ | $+4\lambda_5\ (\lambda_5)^*$ | $+\frac{9{g_2}^4}{4}$ |

**Quantity:** $\lambda_1$

# Terms 3 of 5

| Column | Representative term structure |
|---|---|
| T1 | $+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$ |
| T2 | $-9{g_2}^2\lambda_1$ |
| T3 | $+\frac{3{g_1}^4}{4}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$ | $-9{g_2}^2\lambda_1$ | $+\frac{3{g_1}^4}{4}$ |

**Quantity:** $\lambda_1$

# Terms 4 of 5

| Column | Representative term structure |
|---|---|
| T1 | $+12{\lambda_1}^2$ |
| T2 | $+4{\lambda_{HS_1}^{(1)}}^2$ |
| T3 | $+\frac{{\lambda_{HS_1}^{(A)}}^2}{4}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $+12{\lambda_1}^2$ | $+4{\lambda_{HS_1}^{(1)}}^2$ | $+\frac{{\lambda_{HS_1}^{(A)}}^2}{4}$ |

**Quantity:** $\lambda_1$

# Terms 5 of 5

| Column | Representative term structure |
|---|---|
| T1 | $+4\lambda_{HS_2}^{(1)\ 2}$ |
| T2 | $+\frac{\lambda_{HS_2}^{(A)\ 2}}{4}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 |
|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $+4\lambda_{HS_2}^{(1)\ 2}$ | $+\frac{\lambda_{HS_2}^{(A)\ 2}}{4}$ |

$\beta^{(1)}_{\lambda^{(1)}_{S_1}}$

**Quantity:** $\lambda^{(1)}_{S_1}$

$$16\pi^2\,\mu\frac{d\lambda^{(1)}_{S_1}}{d\mu} = \beta^{(1)}_{\lambda^{(1)}_{S_1}}$$

## Terms 1 of 3

| Column | Representative term structure |
|---|---|
| T1 | $\frac{9{g_2}^4}{4}$ |
| T2 | $+{g_1}^2\left(\frac{3{g_2}^2}{2} - 3\lambda^{(1)}_{S_1}\right)$ |
| T3 | $-9{g_2}^2\lambda^{(1)}_{S_1}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $\frac{9{g_2}^4}{4}$ | $+{g_1}^2\left(\frac{3{g_2}^2}{2} - 3\lambda^{(1)}_{S_1}\right)$ | $-9{g_2}^2\lambda^{(1)}_{S_1}$ |

**Quantity:** $\lambda_{S_1}^{(1)}$

# Terms 2 of 3

| Column | Representative term structure |
|---|---|
| T1 | $+\frac{3{g_1}^4}{4}$ |
| T2 | $+4{\lambda_{12}^{(1)}}^2$ |
| T3 | $+\frac{{\lambda_{12}^{(A)}}^2}{4}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $+\frac{3{g_1}^4}{4}$ | $+4{\lambda_{12}^{(1)}}^2$ | $+\frac{{\lambda_{12}^{(A)}}^2}{4}$ |

**Quantity:** $\lambda_{S_1}^{(1)}$

# Terms 3 of 3

| Column | Representative term structure |
|---|---|
| T1 | $+4{\lambda_{HS_1}^{(1)}}^2$ |
| T2 | $+\frac{{\lambda_{HS_1}^{(A)}}^2}{4}$ |
| T3 | $+12{\lambda_{S_1}^{(1)}}^2$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $+4{\lambda_{HS_1}^{(1)}}^2$ | $+\frac{{\lambda_{HS_1}^{(A)}}^2}{4}$ | $+12{\lambda_{S_1}^{(1)}}^2$ |

$\beta^{(1)}_{\lambda^{(1)}_{S_2}}$

**Quantity:** $\lambda^{(1)}_{S_2}$

$$16\pi^2\,\mu\frac{d\lambda^{(1)}_{S_2}}{d\mu} = \beta^{(1)}_{\lambda^{(1)}_{S_2}}$$

## Terms 1 of 3

| Column | Representative term structure |
|---|---|
| T1 | $\frac{9{g_2}^4}{4}$ |
| T2 | $+{g_1}^2\left(\frac{27{g_2}^2}{2} - 27\lambda^{(1)}_{S_2}\right)$ |
| T3 | $-9{g_2}^2\lambda^{(1)}_{S_2}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $\frac{9{g_2}^4}{4}$ | $+{g_1}^2\left(\frac{27{g_2}^2}{2} - 27\lambda^{(1)}_{S_2}\right)$ | $-9{g_2}^2\lambda^{(1)}_{S_2}$ |

**Quantity:** $\lambda_{S_2}^{(1)}$

# Terms 2 of 3

| Column | Representative term structure |
|---|---|
| T1 | $+\frac{243{g_1}^4}{4}$ |
| T2 | $+4{\lambda_{12}^{(1)}}^2$ |
| T3 | $+\frac{{\lambda_{12}^{(A)}}^2}{4}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $+\frac{243{g_1}^4}{4}$ | $+4{\lambda_{12}^{(1)}}^2$ | $+\frac{{\lambda_{12}^{(A)}}^2}{4}$ |

**Quantity:** $\lambda_{S_2}^{(1)}$

# Terms 3 of 3

| Column | Representative term structure |
|---|---|
| T1 | $+4\lambda_{HS_2}^{(1)\ 2}$ |
| T2 | $+\frac{\lambda_{HS_2}^{(A)\ 2}}{4}$ |
| T3 | $+12\lambda_{S_2}^{(1)\,2}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $+4\lambda_{HS_2}^{(1)\ 2}$ | $+\frac{\lambda_{HS_2}^{(A)\ 2}}{4}$ | $+12\lambda_{S_2}^{(1)\,2}$ |

$\beta^{(1)}_{\lambda^{(1)}_{HS_1}}$

**Quantity:** $\lambda^{(1)}_{HS_1}$

$$16\pi^2\,\mu\frac{d\lambda^{(1)}_{HS_1}}{d\mu} = \beta^{(1)}_{\lambda^{(1)}_{HS_1}}$$

# Terms 1 of 4

| Column | Representative term structure |
|---|---|
| T1 | $\lambda^{(1)}_{HS_1}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ |
| T2 | $+6\lambda_5\ (\lambda_5)^*$ |
| T3 | $+\frac{9{g_2}^4}{4}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $\lambda^{(1)}_{HS_1}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ | $+6\lambda_5\ (\lambda_5)^*$ | $+\frac{9{g_2}^4}{4}$ |

**Quantity:** $\lambda^{(1)}_{HS_1}$

# Terms 2 of 4

| Column | Representative term structure |
|---|---|
| T1 | $-9{g_2}^2\lambda^{(1)}_{HS_1}$ |
| T2 | $+\frac{3{g_1}^4}{4}$ |
| T3 | $-3{g_1}^2\lambda^{(1)}_{HS_1}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $-9{g_2}^2\lambda^{(1)}_{HS_1}$ | $+\frac{3{g_1}^4}{4}$ | $-3{g_1}^2\lambda^{(1)}_{HS_1}$ |

**Quantity:** $\lambda^{(1)}_{HS_1}$

# Terms 3 of 4

| Column | Representative term structure |
|---|---|
| T1 | $+4\lambda^{(1)}_{12}\lambda^{(1)}_{HS_2}$ |
| T2 | $+6\lambda_1\lambda^{(1)}_{HS_1}$ |
| T3 | $+4{\lambda^{(1)}_{HS_1}}^2$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $+4\lambda^{(1)}_{12}\lambda^{(1)}_{HS_2}$ | $+6\lambda_1\lambda^{(1)}_{HS_1}$ | $+4{\lambda^{(1)}_{HS_1}}^2$ |

**Quantity:** $\lambda^{(1)}_{HS_1}$

# Terms 4 of 4

| Column | Representative term structure |
|---|---|
| T1 | $+6\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$ |
| T2 | $+\frac{3\lambda^{(A)\ 2}_{HS_1}}{4}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 |
|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $+6\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$ | $+\frac{3\lambda^{(A)\ 2}_{HS_1}}{4}$ |

$\beta^{(1)}_{\lambda^{(1)}_{HS_2}}$

**Quantity:** $\lambda^{(1)}_{HS_2}$

$$16\pi^2\,\mu\frac{d\lambda^{(1)}_{HS_2}}{d\mu} = \beta^{(1)}_{\lambda^{(1)}_{HS_2}}$$

# Terms 1 of 4

| Column | Representative term structure |
|---|---|
| T1 | $\lambda^{(1)}_{HS_2}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ |
| T2 | $+6\lambda_5\ (\lambda_5)^*$ |
| T3 | $+\frac{9{g_2}^4}{4}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $\lambda^{(1)}_{HS_2}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ | $+6\lambda_5\ (\lambda_5)^*$ | $+\frac{9{g_2}^4}{4}$ |

**Quantity:** $\lambda^{(1)}_{HS_2}$

# Terms 2 of 4

| Column | Representative term structure |
|---|---|
| T1 | $-9{g_2}^2\lambda^{(1)}_{HS_2}$ |
| T2 | $+\frac{27{g_1}^4}{4}$ |
| T3 | $-15{g_1}^2\lambda^{(1)}_{HS_2}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $-9{g_2}^2\lambda^{(1)}_{HS_2}$ | $+\frac{27{g_1}^4}{4}$ | $-15{g_1}^2\lambda^{(1)}_{HS_2}$ |

**Quantity:** $\lambda^{(1)}_{HS_2}$

# Terms 3 of 4

| Column | Representative term structure |
|---|---|
| T1 | $+4\lambda^{(1)}_{12}\lambda^{(1)}_{HS_1}$ |
| T2 | $+6\lambda_1\lambda^{(1)}_{HS_2}$ |
| T3 | $+4{\lambda^{(1)}_{HS_2}}^2$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $+4\lambda^{(1)}_{12}\lambda^{(1)}_{HS_1}$ | $+6\lambda_1\lambda^{(1)}_{HS_2}$ | $+4{\lambda^{(1)}_{HS_2}}^2$ |

**Quantity:** $\lambda_{HS_2}^{(1)}$

# Terms 4 of 4

| Column | Representative term structure |
|---|---|
| T1 | $+6\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$ |
| T2 | $+\frac{3{\lambda_{HS_2}^{(A)}}^2}{4}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 |
|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $+6\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$ | $+\frac{3{\lambda_{HS_2}^{(A)}}^2}{4}$ |

$\beta^{(1)}_{\lambda^{(1)}_{12}}$

**Quantity:** $\lambda^{(1)}_{12}$

$$16\pi^2 \, \mu \frac{d\lambda^{(1)}_{12}}{d\mu} = \beta^{(1)}_{\lambda^{(1)}_{12}}$$

# Terms 1 of 4

| Column | Representative term structure |
|---|---|
| T1 | $3\lambda_5 \, (\lambda_5)^*$ |
| T2 | $+\frac{9{g_2}^4}{4}$ |
| T3 | $-9{g_2}^2\lambda^{(1)}_{12}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $3\lambda_5 \, (\lambda_5)^*$ | $+\frac{9{g_2}^4}{4}$ | $-9{g_2}^2\lambda^{(1)}_{12}$ |

**Quantity:** $\lambda_{12}^{(1)}$

# Terms 2 of 4

| Column | Representative term structure |
|---|---|
| T1 | $+\frac{27{g_1}^4}{4}$ |
| T2 | $-15{g_1}^2\lambda_{12}^{(1)}$ |
| T3 | $+4{\lambda_{12}^{(1)}}^2$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $+\frac{27{g_1}^4}{4}$ | $-15{g_1}^2\lambda_{12}^{(1)}$ | $+4{\lambda_{12}^{(1)}}^2$ |

**Quantity:** $\lambda_{12}^{(1)}$

# Terms 3 of 4

| Column | Representative term structure |
|---|---|
| T1 | $+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$ |
| T2 | $+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$ |
| T3 | $+\frac{3{\lambda_{12}^{(A)}}^2}{4}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$ | $+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$ | $+\frac{3{\lambda_{12}^{(A)}}^2}{4}$ |

**Quantity:** $\lambda_{12}^{(1)}$

# Terms 4 of 4

| Column | Representative term structure |
|---|---|
| T1 | $+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$ |

$\beta^{(1)}_{\lambda_5}$

**Quantity:** $\lambda_5$

$$16\pi^2\,\mu\frac{d\lambda_5}{d\mu} = \beta^{(1)}_{\lambda_5}$$

# Terms 1 of 3

| Column | Representative term structure |
|---|---|
| T1 | $\lambda_5\left(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + \frac{\lambda^{(A)}_{12}}{6} + \frac{2\lambda^{(A)}_{HS_1}}{3} + 2\lambda^{(A)}_{HS_2}\right)$ |
| T2 | $-9{g_2}^2\lambda_5$ |
| T3 | $-9{g_1}^2\lambda_5$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $\lambda_5\left(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + \frac{\lambda^{(A)}_{12}}{6} + \frac{2\lambda^{(A)}_{HS_1}}{3} + 2\lambda^{(A)}_{HS_2}\right)$ | $-9{g_2}^2\lambda_5$ | $-9{g_1}^2\lambda_5$ |

**Quantity:** $\lambda_5$

# Terms 2 of 3

| Column | Representative term structure |
|---|---|
| T1 | $+2\lambda_{12}^{(1)}\lambda_5$ |
| T2 | $+2\lambda_1\lambda_5$ |
| T3 | $+4\lambda_{HS_1}^{(1)}\lambda_5$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $+2\lambda_{12}^{(1)}\lambda_5$ | $+2\lambda_1\lambda_5$ | $+4\lambda_{HS_1}^{(1)}\lambda_5$ |

**Quantity:** $\lambda_5$

# Terms 3 of 3

| Column | Representative term structure |
|---|---|
| T1 | $+4\lambda^{(1)}_{HS_2}\lambda_5$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $+4\lambda^{(1)}_{HS_2}\lambda_5$ |

$\beta^{(1)}_{\lambda_{12}^{(A)}}$

**Quantity:** $\lambda_{12}^{(A)}$

$$16\pi^2\,\mu\frac{d\lambda_{12}^{(A)}}{d\mu} = \beta^{(1)}_{\lambda_{12}^{(A)}}$$

## Terms 1 of 3

| Column | Representative term structure |
|---|---|
| T1 | $\frac{4}{3}\lambda_5\,(\lambda_5)^*$ |
| T2 | $+{g_1}^2\left(18{g_2}^2 - 15\lambda_{12}^{(A)}\right)$ |
| T3 | $-9{g_2}^2\lambda_{12}^{(A)}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $\frac{4}{3}\lambda_5\,(\lambda_5)^*$ | $+{g_1}^2\left(18{g_2}^2 - 15\lambda_{12}^{(A)}\right)$ | $-9{g_2}^2\lambda_{12}^{(A)}$ |

**Quantity:** $\lambda_{12}^{(A)}$

# Terms 2 of 3

| Column | Representative term structure |
|---|---|
| T1 | $+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$ |
| T2 | $+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$ |
| T3 | $+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$ | $+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$ | $+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$ |

**Quantity:** $\lambda_{12}^{(A)}$

# Terms 3 of 3

| Column | Representative term structure |
|---|---|
| T1 | $+\lambda_{HS_1}^{(A)}\,\lambda_{HS_2}^{(A)}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $+\lambda_{HS_1}^{(A)}\,\lambda_{HS_2}^{(A)}$ |

$\beta^{(1)}_{\lambda^{(A)}_{HS_1}}$

**Quantity:** $\lambda^{(A)}_{HS_1}$

$$16\pi^2\,\mu\frac{d\lambda^{(A)}_{HS_1}}{d\mu} = \beta^{(1)}_{\lambda^{(A)}_{HS_1}}$$

# Terms 1 of 3

| Column | Representative term structure |
|---|---|
| T1 | $\lambda^{(A)}_{HS_1}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ |
| T2 | $+\frac{16}{3}\lambda_5\,(\lambda_5)^*$ |
| T3 | $+{g_1}^2\left(6{g_2}^2 - 3\lambda^{(A)}_{HS_1}\right)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $\lambda^{(A)}_{HS_1}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ | $+\frac{16}{3}\lambda_5\,(\lambda_5)^*$ | $+{g_1}^2\left(6{g_2}^2 - 3\lambda^{(A)}_{HS_1}\right)$ |

**Quantity:** $\lambda_{HS_1}^{(A)}$

# Terms 2 of 3

| Column | Representative term structure |
|---|---|
| T1 | $-9{g_2}^2\lambda_{HS_1}^{(A)}$ |
| T2 | $+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$ |
| T3 | $+2\lambda_1\lambda_{HS_1}^{(A)}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $-9{g_2}^2\lambda_{HS_1}^{(A)}$ | $+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$ | $+2\lambda_1\lambda_{HS_1}^{(A)}$ |

**Quantity:** $\lambda^{(A)}_{HS_1}$

# Terms 3 of 3

| Column | Representative term structure |
|---|---|
| T1 | $+8\lambda^{(1)}_{HS_1}\lambda^{(A)}_{HS_1}$ |
| T2 | $+2\lambda^{(A)}_{HS_1}\lambda^{(1)}_{S_1}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 |
|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $+8\lambda^{(1)}_{HS_1}\lambda^{(A)}_{HS_1}$ | $+2\lambda^{(A)}_{HS_1}\lambda^{(1)}_{S_1}$ |

$\beta^{(1)}_{\lambda^{(A)}_{HS_2}}$

**Quantity:** $\lambda^{(A)}_{HS_2}$

$$16\pi^2\,\mu\frac{d\lambda^{(A)}_{HS_2}}{d\mu} = \beta^{(1)}_{\lambda^{(A)}_{HS_2}}$$

# Terms 1 of 3

| Column | Representative term structure |
|---|---|
| T1 | $\lambda^{(A)}_{HS_2}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ |
| T2 | $+16\lambda_5\ (\lambda_5)^*$ |
| T3 | $+{g_1}^2\left(18{g_2}^2 - 15\lambda^{(A)}_{HS_2}\right)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $\lambda^{(A)}_{HS_2}(6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ | $+16\lambda_5\ (\lambda_5)^*$ | $+{g_1}^2\left(18{g_2}^2 - 15\lambda^{(A)}_{HS_2}\right)$ |

**Quantity:** $\lambda_{HS_2}^{(A)}$

# Terms 2 of 3

| Column | Representative term structure |
|---|---|
| T1 | $-9{g_2}^2\lambda_{HS_2}^{(A)}$ |
| T2 | $+\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$ |
| T3 | $+2\lambda_1\lambda_{HS_2}^{(A)}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $-9{g_2}^2\lambda_{HS_2}^{(A)}$ | $+\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$ | $+2\lambda_1\lambda_{HS_2}^{(A)}$ |

**Quantity:** $\lambda_{HS_2}^{(A)}$

# Terms 3 of 3

| Column | Representative term structure |
|---|---|
| T1 | $+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$ |
| T2 | $+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 |
|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$ | $+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$ |

# Dimension-five $\psi^2\phi^2$ Wilson coefficients

The Wilson tensor is reported in the real-scalar component basis $C_{ijab}$. Each component with a nonzero one-loop beta function has its own table. Components absent at the matching boundary but generated by operator mixing are included automatically.

## $\beta^{(1)}_{C_{1133}}$

**Quantity:** 1, 1, 3, 3

$$16\pi^2\,\mu\frac{dC_{1133}}{d\mu} = \beta^{(1)}_{C_{1133}}$$

| Column | Representative term structure |
|---|---|
| T1 | $\frac{lambdaT_3\overline{y_1}\,\overline{y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $-\frac{2lambdaT_3\overline{y_1}\,\overline{y_2}}{3MF}$ |

$\beta^{(1)}_{C_{1134}}$

**Quantity:** 1, 1, 3, 4

$$16\pi^2\,\mu\frac{dC_{1134}}{d\mu} = \beta^{(1)}_{C_{1134}}$$

| Column | Representative term structure |
|---|---|
| T1 | $\frac{ilambdaT_3\overline{y_1}\,\overline{y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $-\frac{2ilambdaT_3\overline{y_1}\,\overline{y_2}}{3MF}$ |

$\beta^{(1)}_{C_{1144}}$

**Quantity:** 1, 1, 4, 4

$$16\pi^2\,\mu\frac{dC_{1144}}{d\mu} = \beta^{(1)}_{C_{1144}}$$

| Column | Representative term structure |
|---|---|
| T1 | $\frac{lambdaT_3\overline{y_1\,y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $\frac{2lambdaT_3\overline{y_1\,y_2}}{3MF}$ |

$\beta^{(1)}_{C_{11511}}$

**Quantity:** 1, 1, 5, 11

$$16\pi^2\,\mu\,\frac{dC_{11511}}{d\mu} = \beta^{(1)}_{C_{11511}}$$

# Terms 1 of 2

| Column | Representative term structure |
|---|---|
| T1 | $\frac{lambda12Inv2\overline{y_1\,y_2}}{MF}$ |
| T2 | $\frac{\sqrt{3}g_1^2\overline{y_1\,y_2}}{MF}$ |
| T3 | $\frac{\sqrt{3}g_2^2\overline{y_1\,y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $\frac{lambda12Inv2\overline{y_1\,y_2}}{3MF}$ | $\frac{\sqrt{3}g_1^2\overline{y_1\,y_2}}{2MF}$ | $\frac{\sqrt{3}g_2^2\overline{y_1\,y_2}}{4MF}$ |

**Quantity:** 1, 1, 5, 11

# Terms 2 of 2

| Column | Representative term structure |
|---|---|
| T1 | $\frac{\sqrt{3}ye\overline{y_1 y_2 ye}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $-\frac{\sqrt{3}ye\overline{y_1 y_2 ye}}{12MF}$ |

$\beta^{(1)}_{C_{11512}}$

**Quantity:** 1, 1, 5, 12

$$16\pi^2\,\mu\,\frac{dC_{11512}}{d\mu} = \beta^{(1)}_{C_{11512}}$$

# Terms 1 of 2

| Column | Representative term structure |
|---|---|
| T1 | $\frac{ilambda12Inv2\overline{y_1\,y_2}}{MF}$ |
| T2 | $\frac{\sqrt{3}ig_1^2\overline{y_1\,y_2}}{MF}$ |
| T3 | $\frac{\sqrt{3}ig_2^2\overline{y_1\,y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $\frac{ilambda12Inv2\overline{y_1y_2}}{3MF}$ | $\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{2MF}$ | $\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{4MF}$ |

**Quantity:** 1, 1, 5, 12

# Terms 2 of 2

| Column | Representative term structure |
|---|---|
| T1 | $\frac{\sqrt{3}iye\overline{y_1 y_2 ye}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $-\frac{\sqrt{3}iye\overline{y_1 y_2 ye}}{12MF}$ |

$\beta^{(1)}_{C_{11611}}$

**Quantity:** 1, 1, 6, 11

$$16\pi^2\,\mu\,\frac{dC_{11611}}{d\mu} = \beta^{(1)}_{C_{11611}}$$

# Terms 1 of 2

| Column | Representative term structure |
|---|---|
| T1 | $\frac{ilambda12Inv2\overline{y_1\,y_2}}{MF}$ |
| T2 | $\frac{\sqrt{3}ig_1^2\overline{y_1\,y_2}}{MF}$ |
| T3 | $\frac{\sqrt{3}ig_2^2\overline{y_1\,y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $-\frac{ilambda12Inv2\overline{y_1\,y_2}}{3MF}$ | $-\frac{\sqrt{3}ig_1^2\overline{y_1\,y_2}}{2MF}$ | $-\frac{\sqrt{3}ig_2^2\overline{y_1\,y_2}}{4MF}$ |

**Quantity:** 1, 1, 6, 11

# Terms 2 of 2

| Column | Representative term structure |
|---|---|
| T1 | $\frac{\sqrt{3}iye\overline{y_1 y_2 ye}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $\frac{\sqrt{3}iye\overline{y_1 y_2 ye}}{12MF}$ |

$\beta^{(1)}_{C_{11612}}$

**Quantity:** 1, 1, 6, 12

$$16\pi^2\,\mu\,\frac{dC_{11612}}{d\mu} = \beta^{(1)}_{C_{11612}}$$

# Terms 1 of 2

| Column | Representative term structure |
|---|---|
| T1 | $\frac{lambda12Inv2\overline{y_1\,y_2}}{MF}$ |
| T2 | $\frac{\sqrt{3}g_1^2\overline{y_1\,y_2}}{MF}$ |
| T3 | $\frac{\sqrt{3}g_2^2\overline{y_1\,y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $\frac{lambda12Inv2\overline{y_1\,y_2}}{3MF}$ | $\frac{\sqrt{3}g_1^2\overline{y_1\,y_2}}{2MF}$ | $\frac{\sqrt{3}g_2^2\overline{y_1\,y_2}}{4MF}$ |

**Quantity:** 1, 1, 6, 12

# Terms 2 of 2

| Column | Representative term structure |
|---|---|
| T1 | $\frac{\sqrt{3}ye\overline{y_1 y_2 ye}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $-\frac{\sqrt{3}ye\overline{y_1 y_2 ye}}{12MF}$ |

$\beta^{(1)}_{C_{1213}}$

**Quantity:** 1, 2, 1, 3

$$16\pi^2\,\mu\frac{dC_{1213}}{d\mu} = \beta^{(1)}_{C_{1213}}$$

| Column | Representative term structure |
|---|---|
| T1 | $\frac{lambdaT_3\overline{y_1\,y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $\frac{lambdaT_3\overline{y_1\,y_2}}{3MF}$ |

$\beta^{(1)}_{C_{1214}}$

**Quantity:** 1, 2, 1, 4

$$16\pi^2\,\mu\frac{dC_{1214}}{d\mu} = \beta^{(1)}_{C_{1214}}$$

| Column | Representative term structure |
|---|---|
| T1 | $\frac{ilambdaT_3\overline{y_1 y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $\frac{ilambdaT_3\overline{y_1 y_2}}{3MF}$ |

$\beta^{(1)}_{C_{1223}}$

**Quantity:** 1, 2, 2, 3

$$16\pi^2\,\mu\frac{dC_{1223}}{d\mu} = \beta^{(1)}_{C_{1223}}$$

| Column | Representative term structure |
|---|---|
| T1 | $\frac{ilambdaT_3\overline{y_1 y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $\frac{ilambdaT_3\overline{y_1 y_2}}{3MF}$ |

$\beta^{(1)}_{C_{1224}}$

**Quantity:** 1, 2, 2, 4

$$16\pi^2\,\mu\frac{dC_{1224}}{d\mu} = \beta^{(1)}_{C_{1224}}$$

| Column | Representative term structure |
|---|---|
| T1 | $\frac{lambdaT_3\overline{y_1\,y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $-\frac{lambdaT_3\overline{y_1\,y_2}}{3MF}$ |

$\beta^{(1)}_{C_{1259}}$

**Quantity:** 1, 2, 5, 9

$$16\pi^2\,\mu\frac{dC_{1259}}{d\mu} = \beta^{(1)}_{C_{1259}}$$

# Terms 1 of 2

| Column | Representative term structure |
|---|---|
| T1 | $\frac{lambda12Inv2\overline{y_1\,y_2}}{MF}$ |
| T2 | $\frac{\sqrt{3}g_1^2\overline{y_1\,y_2}}{MF}$ |
| T3 | $\frac{\sqrt{3}g_2^2\overline{y_1\,y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $-\frac{lambda12Inv2\overline{y_1\,y_2}}{6MF}$ | $-\frac{\sqrt{3}g_1^2\overline{y_1\,y_2}}{4MF}$ | $-\frac{\sqrt{3}g_2^2\overline{y_1\,y_2}}{8MF}$ |

**Quantity:** 1, 2, 5, 9

# Terms 2 of 2

| Column | Representative term structure |
|---|---|
| T1 | $\frac{\sqrt{3}ye\overline{y_1 y_2 ye}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $\frac{\sqrt{3}ye\overline{y_1 y_2 ye}}{24MF}$ |

$\beta^{(1)}_{C_{12510}}$

**Quantity:** 1, 2, 5, 10

$$16\pi^2\,\mu\,\frac{dC_{12510}}{d\mu} = \beta^{(1)}_{C_{12510}}$$

# Terms 1 of 2

| Column | Representative term structure |
|---|---|
| T1 | $\frac{ilambda12Inv2\overline{y_1\,y_2}}{MF}$ |
| T2 | $\frac{\sqrt{3}ig_1^2\overline{y_1\,y_2}}{MF}$ |
| T3 | $\frac{\sqrt{3}ig_2^2\overline{y_1\,y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $-\frac{ilambda12Inv2\overline{y_1\,y_2}}{6MF}$ | $-\frac{\sqrt{3}ig_1^2\overline{y_1\,y_2}}{4MF}$ | $-\frac{\sqrt{3}ig_2^2\overline{y_1\,y_2}}{8MF}$ |

**Quantity:** 1, 2, 5, 10

# Terms 2 of 2

| Column | Representative term structure |
|---|---|
| T1 | $\frac{\sqrt{3}iye\overline{y_1 y_2 ye}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $\frac{\sqrt{3}iye\overline{y_1 y_2 ye}}{24MF}$ |

$\beta^{(1)}_{C_{1269}}$

**Quantity:** 1, 2, 6, 9

$$16\pi^2\,\mu\frac{dC_{1269}}{d\mu} = \beta^{(1)}_{C_{1269}}$$

# Terms 1 of 2

| Column | Representative term structure |
|---|---|
| T1 | $\frac{ilambda12Inv2\overline{y_1\,y_2}}{MF}$ |
| T2 | $\frac{\sqrt{3}ig_1^2\overline{y_1\,y_2}}{MF}$ |
| T3 | $\frac{\sqrt{3}ig_2^2\overline{y_1\,y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $\frac{ilambda12Inv2\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{4MF}$ | $\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{8MF}$ |

**Quantity:** 1, 2, 6, 9

# Terms 2 of 2

| Column | Representative term structure |
|---|---|
| T1 | $\frac{\sqrt{3}iye\overline{y_1 y_2 ye}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $-\frac{\sqrt{3}iye\overline{y_1 y_2 ye}}{24MF}$ |

$\beta^{(1)}_{C_{12610}}$

**Quantity:** 1, 2, 6, 10

$$16\pi^2\,\mu\,\frac{dC_{12610}}{d\mu} = \beta^{(1)}_{C_{12610}}$$

# Terms 1 of 2

| Column | Representative term structure |
|---|---|
| T1 | $\frac{lambda12Inv2\overline{y_1\,y_2}}{MF}$ |
| T2 | $\frac{\sqrt{3}g_1^2\overline{y_1\,y_2}}{MF}$ |
| T3 | $\frac{\sqrt{3}g_2^2\overline{y_1\,y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $-\frac{lambda12Inv2\overline{y_1\,y_2}}{6MF}$ | $-\frac{\sqrt{3}g_1^2\overline{y_1\,y_2}}{4MF}$ | $-\frac{\sqrt{3}g_2^2\overline{y_1\,y_2}}{8MF}$ |

**Quantity:** 1, 2, 6, 10

# Terms 2 of 2

| Column | Representative term structure |
|---|---|
| T1 | $\frac{\sqrt{3}ye\overline{y_1 y_2 ye}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $\frac{\sqrt{3}ye\overline{y_1 y_2 ye}}{24MF}$ |

$\beta^{(1)}_{C_{12711}}$

**Quantity:** 1, 2, 7, 11

$$16\pi^2\,\mu\,\frac{dC_{12711}}{d\mu} = \beta^{(1)}_{C_{12711}}$$

# Terms 1 of 2

| Column | Representative term structure |
|---|---|
| T1 | $\frac{lambda12Inv2\overline{y_1\,y_2}}{MF}$ |
| T2 | $\frac{\sqrt{3}g_1^2\overline{y_1\,y_2}}{MF}$ |
| T3 | $\frac{\sqrt{3}g_2^2\overline{y_1\,y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $\frac{lambda12Inv2\overline{y_1\,y_2}}{6MF}$ | $\frac{\sqrt{3}g_1^2\overline{y_1\,y_2}}{4MF}$ | $\frac{\sqrt{3}g_2^2\overline{y_1\,y_2}}{8MF}$ |

**Quantity:** 1, 2, 7, 11

# Terms 2 of 2

| Column | Representative term structure |
|---|---|
| T1 | $\frac{\sqrt{3}ye\overline{y_1 y_2 ye}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $-\frac{\sqrt{3}ye\overline{y_1 y_2 ye}}{24MF}$ |

$\beta^{(1)}_{C_{12712}}$

**Quantity:** 1, 2, 7, 12

$$16\pi^2\,\mu\,\frac{dC_{12712}}{d\mu} = \beta^{(1)}_{C_{12712}}$$

# Terms 1 of 2

| Column | Representative term structure |
|---|---|
| T1 | $\frac{ilambda12Inv2\overline{y_1\,y_2}}{MF}$ |
| T2 | $\frac{\sqrt{3}ig_1^2\overline{y_1\,y_2}}{MF}$ |
| T3 | $\frac{\sqrt{3}ig_2^2\overline{y_1\,y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $\frac{ilambda12Inv2\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{4MF}$ | $\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{8MF}$ |

**Quantity:** 1, 2, 7, 12

# Terms 2 of 2

| Column | Representative term structure |
|---|---|
| T1 | $\frac{\sqrt{3}iye\overline{y_1 y_2 ye}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $-\frac{\sqrt{3}iye\overline{y_1 y_2 ye}}{24MF}$ |

$\beta^{(1)}_{C_{12811}}$

**Quantity:** 1, 2, 8, 11

$$16\pi^2\,\mu\,\frac{dC_{12811}}{d\mu} = \beta^{(1)}_{C_{12811}}$$

# Terms 1 of 2

| Column | Representative term structure |
|---|---|
| T1 | $\frac{ilambda12Inv2\overline{y_1\,y_2}}{MF}$ |
| T2 | $\frac{\sqrt{3}ig_1^2\overline{y_1\,y_2}}{MF}$ |
| T3 | $\frac{\sqrt{3}ig_2^2\overline{y_1\,y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $-\frac{ilambda12Inv2\overline{y_1\,y_2}}{6MF}$ | $-\frac{\sqrt{3}ig_1^2\overline{y_1\,y_2}}{4MF}$ | $-\frac{\sqrt{3}ig_2^2\overline{y_1\,y_2}}{8MF}$ |

**Quantity:** 1, 2, 8, 11

# Terms 2 of 2

| Column | Representative term structure |
|---|---|
| T1 | $\frac{\sqrt{3}iye\overline{y_1 y_2 ye}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $\frac{\sqrt{3}iye\overline{y_1 y_2 ye}}{24MF}$ |

$\beta^{(1)}_{C_{12812}}$

**Quantity:** 1, 2, 8, 12

$$16\pi^2\,\mu\,\frac{dC_{12812}}{d\mu} = \beta^{(1)}_{C_{12812}}$$

# Terms 1 of 2

| Column | Representative term structure |
|---|---|
| T1 | $\frac{lambda12Inv2\overline{y_1\,y_2}}{MF}$ |
| T2 | $\frac{\sqrt{3}g_1^2\overline{y_1\,y_2}}{MF}$ |
| T3 | $\frac{\sqrt{3}g_2^2\overline{y_1\,y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $\frac{lambda12Inv2\overline{y_1\,y_2}}{6MF}$ | $\frac{\sqrt{3}g_1^2\overline{y_1\,y_2}}{4MF}$ | $\frac{\sqrt{3}g_2^2\overline{y_1\,y_2}}{8MF}$ |

**Quantity:** 1, 2, 8, 12

# Terms 2 of 2

| Column | Representative term structure |
|---|---|
| T1 | $\frac{\sqrt{3}ye\overline{y_1 y_2 ye}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $-\frac{\sqrt{3}ye\overline{y_1 y_2 ye}}{24MF}$ |

$\beta^{(1)}_{C_{2211}}$

**Quantity:** 2, 2, 1, 1

$$16\pi^2\,\mu\frac{dC_{2211}}{d\mu} = \beta^{(1)}_{C_{2211}}$$

| Column | Representative term structure |
|---|---|
| T1 | $\frac{lambdaT_3\overline{y_1\,y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $-\frac{2lambdaT_3\overline{y_1\,y_2}}{3MF}$ |

$\beta^{(1)}_{C_{2212}}$

**Quantity:** 2, 2, 1, 2

$$16\pi^2\,\mu\frac{dC_{2212}}{d\mu} = \beta^{(1)}_{C_{2212}}$$

| Column | Representative term structure |
|---|---|
| T1 | $\frac{ilambdaT_3\overline{y_1}\,\overline{y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $-\frac{2ilambdaT_3\overline{y_1}\,\overline{y_2}}{3MF}$ |

$\beta^{(1)}_{C_{2222}}$

**Quantity:** 2, 2, 2, 2

$$16\pi^2\,\mu\frac{dC_{2222}}{d\mu} = \beta^{(1)}_{C_{2222}}$$

| Column | Representative term structure |
|---|---|
| T1 | $\frac{lambdaT_3\overline{y_1\,y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $\frac{2lambdaT_3\overline{y_1\,y_2}}{3MF}$ |

$\beta^{(1)}_{C_{2279}}$

**Quantity:** 2, 2, 7, 9

$$16\pi^2\,\mu\frac{dC_{2279}}{d\mu} = \beta^{(1)}_{C_{2279}}$$

# Terms 1 of 2

| Column | Representative term structure |
|---|---|
| T1 | $\frac{lambda12Inv2\overline{y_1\,y_2}}{MF}$ |
| T2 | $\frac{\sqrt{3}g_1^2\overline{y_1\,y_2}}{MF}$ |
| T3 | $\frac{\sqrt{3}g_2^2\overline{y_1\,y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $-\frac{lambda12Inv2\overline{y_1\,y_2}}{3MF}$ | $-\frac{\sqrt{3}g_1^2\overline{y_1\,y_2}}{2MF}$ | $-\frac{\sqrt{3}g_2^2\overline{y_1\,y_2}}{4MF}$ |

**Quantity:** 2, 2, 7, 9

# Terms 2 of 2

| Column | Representative term structure |
|---|---|
| T1 | $\frac{\sqrt{3}ye\overline{y_1 y_2 ye}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $\frac{\sqrt{3}ye\overline{y_1 y_2 ye}}{12MF}$ |

$\beta^{(1)}_{C_{22710}}$

**Quantity:** 2, 2, 7, 10

$$16\pi^2\,\mu\,\frac{dC_{22710}}{d\mu} = \beta^{(1)}_{C_{22710}}$$

# Terms 1 of 2

| Column | Representative term structure |
|---|---|
| T1 | $\frac{ilambda12Inv2\overline{y_1\,y_2}}{MF}$ |
| T2 | $\frac{\sqrt{3}ig_1^2\overline{y_1\,y_2}}{MF}$ |
| T3 | $\frac{\sqrt{3}ig_2^2\overline{y_1\,y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $-\frac{ilambda12Inv2\overline{y_1\,y_2}}{3MF}$ | $-\frac{\sqrt{3}ig_1^2\overline{y_1\,y_2}}{2MF}$ | $-\frac{\sqrt{3}ig_2^2\overline{y_1\,y_2}}{4MF}$ |

**Quantity:** 2, 2, 7, 10

# Terms 2 of 2

| Column | Representative term structure |
|---|---|
| T1 | $\frac{\sqrt{3}iye\overline{y_1 y_2 ye}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $\frac{\sqrt{3}iye\overline{y_1 y_2 ye}}{12MF}$ |

$\beta^{(1)}_{C_{2289}}$

**Quantity:** 2, 2, 8, 9

$$16\pi^2\,\mu\frac{dC_{2289}}{d\mu} = \beta^{(1)}_{C_{2289}}$$

# Terms 1 of 2

| Column | Representative term structure |
|---|---|
| T1 | $\frac{ilambda12Inv2\overline{y_1\,y_2}}{MF}$ |
| T2 | $\frac{\sqrt{3}ig_1^2\overline{y_1\,y_2}}{MF}$ |
| T3 | $\frac{\sqrt{3}ig_2^2\overline{y_1\,y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $\frac{ilambda12Inv2\overline{y_1y_2}}{3MF}$ | $\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{2MF}$ | $\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{4MF}$ |

**Quantity:** 2, 2, 8, 9

# Terms 2 of 2

| Column | Representative term structure |
|---|---|
| T1 | $\frac{\sqrt{3}iye\overline{y_1 y_2 ye}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $-\frac{\sqrt{3}iye\overline{y_1 y_2 ye}}{12MF}$ |

$\beta^{(1)}_{C_{22810}}$

**Quantity:** 2, 2, 8, 10

$$16\pi^2\,\mu\,\frac{dC_{22810}}{d\mu} = \beta^{(1)}_{C_{22810}}$$

# Terms 1 of 2

| Column | Representative term structure |
|---|---|
| T1 | $\frac{lambda12Inv2\overline{y_1\,y_2}}{MF}$ |
| T2 | $\frac{\sqrt{3}g_1^2\overline{y_1\,y_2}}{MF}$ |
| T3 | $\frac{\sqrt{3}g_2^2\overline{y_1\,y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $-\frac{lambda12Inv2\overline{y_1\,y_2}}{3MF}$ | $-\frac{\sqrt{3}g_1^2\overline{y_1\,y_2}}{2MF}$ | $-\frac{\sqrt{3}g_2^2\overline{y_1\,y_2}}{4MF}$ |

**Quantity:** 2, 2, 8, 10

# Terms 2 of 2

| Column | Representative term structure |
|---|---|
| T1 | $\frac{\sqrt{3}ye\overline{y_1 y_2 ye}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-C | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $\frac{\sqrt{3}ye\overline{y_1 y_2 ye}}{12MF}$ |